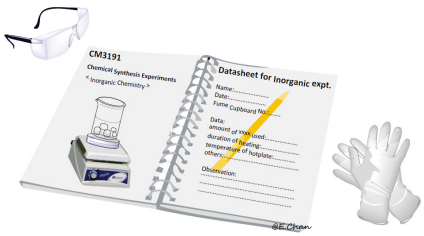
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AY24/25 Semester 1

CM3191 (INORGANIC part)

Chemical Synthesis Experiments



Dr. Edith Chan

NUS National University of Singapore

Disclaimer: All materials, information, weblinks and pictures contained in this set of notes are for educational purposes only.

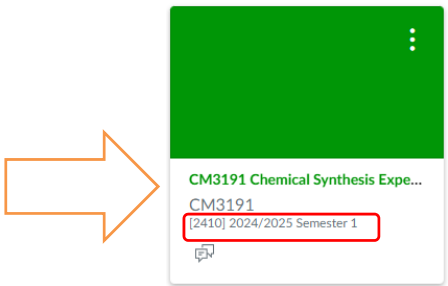
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INORGANIC Lecturer-in-charge: Contact Information

Dr. Edith S. H. CHAN
Email: e.chan@nus.edu.sg

Please use your NUS email account to send any query!

Course Materials (Inorganic): Canvas - CM3191 Chemical Synthesis Experiments
(refer to https://canvas.nus.edu.sg/)



**** For lab class (Mon./Thu.) registration, you have to do it via CourseReg on 12-Aug-2024 (refer to Canvas-announcement posted on 7-Aug-2024). Otherwise, you may not be allowed to take CM3191.****

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CM3191 Instructors & Assessments

- Inorganic Chemistry (Lectures, Monday and Thursday Lab Sessions) & **Course Coordinator**: Dr. Edith S. H. Chan
- Organic Chemistry (Lectures): Dr. Zhang Sheng
- Organic Chemistry (Lab Sessions): Dr. Zhang Sheng & Dr. Hoang Truong Giang

	Inorganic	Organic
Risk Assessment(s) / Safety Test	5%	5%
Lab Reports	30%	30%
Closed-book Written Test	15%	15%
Total % of CAs in each part	50%	50%

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IMPORTANT NOTICES

During the "hands-on" sessions in laboratory, ALL students, teaching assistants and staff MUST wear safety googles and lab coats in the laboratory.

- Please take note on the safe-distancing during practical classes, for safety reasons.
- If you feel sick or have medical reasons, please do NOT come to campus and seek medical advices from medical doctors. You can contact the lecturer-in-charge by email for the makeup lab class such reason, with the provision of official medical certificate.

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Laboratory Safety

- Attire (full-length pants, covered shoes)
- PPE: lab coat (prepared by your own), safety goggles (prepared by your own), gloves (provided in the lab)
- No three-quarter pants
- No-show socks should not be worn (wear high-cut socks)
- No high heels, open-toe shoes or ballet flats
- No dangling jewelry
- Long hair should be tied or bundled
- No eating and drinking is allowed inside the lab
- ***Skinny* jeans/pants/leggings or Yoga pants should be avoided.**

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Laboratory Safety: Disposable Gloves

- In our teaching laboratory, two types of disposable gloves, nitrile gloves and latex gloves will be provided.
- Nitrile gloves are made of synthetic rubber materials and thus they offer resistance to the chemicals such as hexane, acid, alcohol oil and, etc. The breakthrough time is about 4-minute when chloroform or dichloromethane or acetone is used.
- When acetone is used, the gloves made of latex (also known as natural rubber) has about a 10-minute breakthrough time that is a bit better than nitrile gloves, but latex degrades in oil. The best choice to handle acetone is butyl rubber gloves. As butyl rubber is an artificial polymer and more specialized than most glove materials, butyl rubber gloves are more expensive.
- Breakthrough time (BT) is defined as the elapsed time between initial contact of the liquid chemical with the outside surface of the glove and the time at which the permeation rate reaches 0.1 $mg/m^2/sec$. The adequate protection is no longer provided by the gloves when breakthrough occurs.
- Permeation rate (PR) is the measured steady state flow of the permeating chemical through the glove elastomer. Glove thickness is one of the parameters to determine the resistance to permeation.

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Laboratory Safety: Latex Allergies & Symptoms

- IMPORTANCE: Some people have latex allergies and cannot wear latex gloves. Latex allergy symptoms may include hives, itching, stuffy or runny nose. It can cause asthma symptoms of wheezing, chest tightness and difficulty breathing. Symptoms start within minutes after exposure to latex containing products. Allergic skin problems can occur following direct contact with allergic latex proteins in latex glove products. Symptoms may include immediate itching, redness and swelling of skin that touched the item containing latex. A second type of skin allergy called "allergic contact dermatitis", which is recognized by the eczema and blisters on the back of the hands, may be caused by chemicals used to manufacture rubber gloves. It is similar to a poison ivy rash, that begins 1 to 3 days after wearing rubber gloves. (VISIT https://acaai.org/allergies/allergic-conditions/latex-allergy/ FOR MORE DETAILS)



Hand dermatitis that developed from wearing latex gloves

- It is also important to note that as of January 2017, powdered (latex or other types) gloves are not used in medical applications. They are associated with health risks, including severe airway inflammation, wound inflammation, and post-surgical adhesions.

Source: https://acaai.org/allergies/allergic-conditions/latex-allergy/

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List of INORGANIC experiments

INORGANIC Experiment 1

Homogenous Catalysis: Tris(triphenylphosphine)ruthenium(II) Dichloride – Synthesis, Characterization and Catalytic Studies

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$

Session II: Controlled Study - Oxidation of 1-phenylethanol to acetophenone in the absence of catalyst

Session III: Study of Catalytic Reaction: $RuCl_2(PPh_3)_3$–catalysed Oxidation of 1-phenylethanol to acetophenone

INORGANIC Experiment 2

Metallocene: Ferrocene - Synthesis and Characterization

- Session I: Synthesis and Microscopic Image of Crude Ferrocene
- Session II: Purification and Characterization of Ferrocene

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Lecture & Practical: Arrangement

Note: Announcement will be given in the 1st lecture of each part if there is any change.

- Lecture schedule: Monday (8:00 a.m. – 10:00 a.m.) at LT26
- Lab class schedule: Monday or Thursday (10:00 a.m. – 4:00 p.m.) at S5-LEVEL4 Chemical Synthesis Teaching Lab.

Week	Lecture	Monday Lab	Thursday Lab
1	**Inorganic**	*No Class*	*No Class*
2	**Inorganic**	**Inorganic**	**Inorganic**
3	**Inorganic**	**Inorganic**	**Inorganic**
4	**Inorganic**	**Inorganic**	**Inorganic**
5	**Inorganic**	**Inorganic**	**Inorganic**
6	**Organic**	**Inorganic**	**Inorganic**
Reading week		*No Class*	
7	**Organic**	**Organic**	**Organic**
8	**Organic**	**Organic**	**Organic**
9	**Organic**	**Organic**	**Organic**
10	**Organic**	**Organic**	**Organic**
11	*No Class (tentative)**	**Organic**	*No Class* (Deepavali)
12	*No Class (tentative)**	*No Class*	**Organic**
13	**Written Test** (for both inorganic and organic parts)	*No Class (for makeup lab only)*	*No Class (for makeup lab only)*

* Remarks: For organic part, please check with Dr. Zhang Sheng.

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Inorganic Practical Schedule

Note: No lab class in week 1.

Lab schedule	Week 2 (Mon./Thu.)	Week 3 (Mon./Thu.)	Week 4 (Mon./Thu.)	Week 5 (Mon./Thu.)	Week 6 (Mon./Thu.)
ALL Students	Expt. 1 (session I)	Expt. 1 (session II)	Expt. 1 (session III)	Expt. 2 (session I)	Expt. 2 (session II)
Instrument(s) / technique used	IR & ESI-MS	1H-NMR, use of micropipette	1H-NMR, use of micropipette	Stereomicroscope, use of syringe	Stereomicroscope, magnetic susceptibility balance, IR, UV-vis & ESI-MS, 1H-NMR

**** For lab class (Mon./Thu.) registration, you have to do it via CourseReg on 12-Aug-2024 (refer to Canvas-announcement posted on 7-Aug-2024). Otherwise, you may not be allowed to take CM3191.****

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Inorganic Experiments – Journal Search

- For the lab manual of each inorganic experiment, there is a list of references which are available in NUS library. For the abbreviated journal name which may not be recognized by NUS lib. system, its full name can be searched by search engine such as google search.

- **For Journal Search,**

(i) *go to* http://www.nus.edu.sg
(ii) *click* "Libraries"
(iii) *type* the title of journal in "search"
(iv) *Click "* *Full Text Online* *" & browse* by issue

e.g. E. Wong, C. M. Giandomenico, *Chem. Rev*. **1999**, *99*, 2451-2466.

Author(s) | Title of journal | year | vol. no. | page no.

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Inorganic Lectures – Aims & Topics

< Note: No recording of lectures ! >

- Aims
- To provide some basic knowledge on the risk assessment, specific techniques or topics which are covered in CM3191 inorganic experiments

- Topics
- Brief introduction to Transition Metals
- ESI-MS in inorganic chemistry
- ^{1}H-NMR & ^{31}P-NMR in inorganic chemistry
- Magnetic Susceptibility
- CBC Electron Counting for (mononuclear) organometallic compounds
- Other topics for inorganic practical

Week 1-4: Inorganic Practical Arrangement

Week	Lecture [LT26] (Monday, 8:00 a.m. – 10:00 a.m.) < Note: No recording of lectures ! >	Practical (Mon.) [S5-04] (10am – 4pm)	Practical (Thu.) [S5-04] (10am – 4pm)	Assignment/Activity for CM3191-INORGANIC
	Complete the "CM3191 Undergraduate Safety Quiz" via Canvas-Assignment by 23:59 pm on 16-Aug-2024 (Week 1, Fri.).			
1	Inorganic lecture	*No Lab class*	*No Lab class*	(i) Attend inorganic lecture (ii) **Register the lab class (Mon./Thu.) via CourseReg on 12-Aug-2024 (Week 1, Mon.)** (iii) **Complete the "CM3191 Undergraduate Safety Quiz" via Canvas-Assignment by 23:59 pm on 16-Aug-2024 (Week 1, Fri.).**
2	Inorganic lecture	Expt. 1 (session I)	Expt. 1 (session I)	Attend inorganic lecture & lab class
3	Inorganic lecture	Expt. 1 (session II)	Expt. 1 (session II)	(i) Attend inorganic lecture & lab class (ii) Prepare inorganic risk assessment
4	Inorganic lecture	Expt. 1 (session III)	Expt. 1 (session III)	(i) Attend inorganic lecture & lab class **(ii)** **[ALL groups]** Prepare risk assessment for inorganic expt. 2 (session II) and submit it via Canvas-Assignments by 10:00 a.m. on 16-Sep-2024 (Mon.; week 6). **[5% CA]**

Week 5-7: Inorganic Practical Arrangement

Week	Lecture [LT26] (Monday, 8:00 a.m. – 10:00 a.m.) < Note: No recording of lectures ! >	Practical (Mon.) (10am – 4pm)	Practical (Thu.) (10am – 4pm)	Assignment/Activity for CM3191-INORGANIC
5	Inorganic lecture	Expt. 2 (session I)	Expt. 2 (session I)	(i) Attend inorganic lecture & lab class **(ii)** **[ALL groups]** Prepare risk assessment for inorganic expt. 2 (session II) and submit it via Canvas-Assignments by 10:00 a.m. on 16-Sep-2024 (Mon.; week 6). **[5% CA]**
6	Organic*	Expt. 2 (session II)	Expt. 2 (session II)	(i) Attend lab class (ii) Prepare inorganic lab reports
Recess week	*No class*			
7	Organic	Organic	Organic	(i) Prepare inorganic lab reports

**Remarks: For organic part, please contact Dr. Zhang Sheng.*

Week 8-13: Inorganic Practical Arrangement

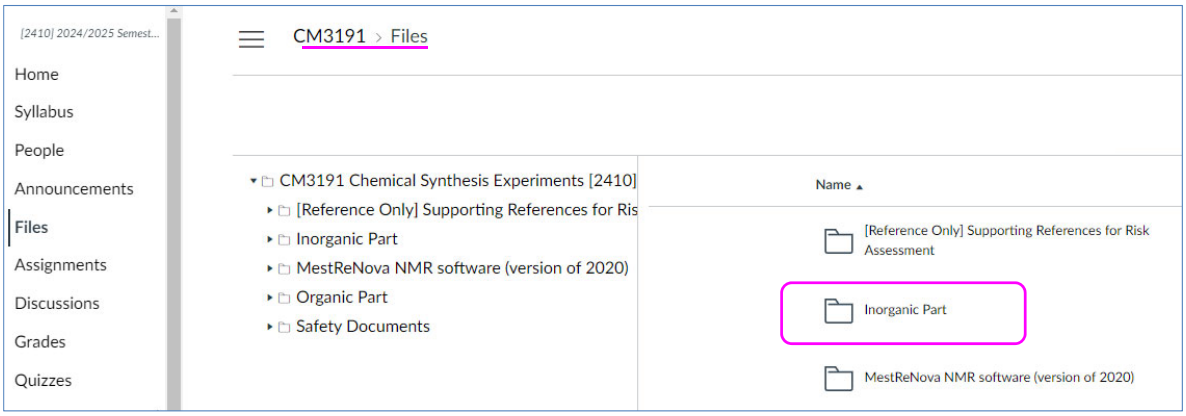
Week	Lecture [LT26] (Monday, 8:00 a.m. – 10:00 a.m.) < Note: No recording of lectures ! >	Practical (Mon. group) (10am – 4pm)	Practical (Thu. group) (10am – 4pm)	Assignment/Activity for CM3191-INORGANIC
8 – 9*	Organic	Organic	Organic	Prepare inorganic lab reports
10*	Organic	Organic	Organic	Prepare inorganic lab reports
11*	*No lecture (tentative)**	*Organic*	*No Lab class (Deepavali)*	Prepare inorganic lab reports
12	*No lecture (tentative)**	*No Lab class*	Organic	**[ALL Groups]** Submit **inorganic expt. 1 proforma** via Canvas-Assignments by 10:00 a.m. on 8-Nov-2024 (Fri.; week 12). **[15% CA]** **[ALL Groups]** Submit **inorganic expt. 2 short report** via Canvas-Assignments by 10:00 a.m. on 8-Nov-2024 (Fri.; week 12). **[15% CA]**
13	**Closed-Book Written Test** (both inorganic & organic parts)	*No lab class*	*No lab class*	Take the written test as scheduled. **[15% CA for inorganic]**

**Remarks: For organic part and the lecture slots in week 11-12, please contact Dr. Zhang Sheng.*

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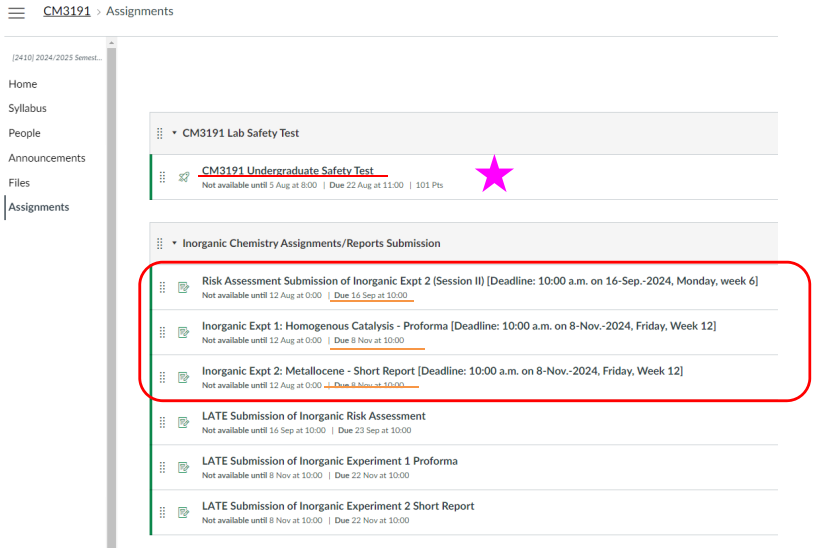
Files for Inorganic part

- Available NOW



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Undergraduate Safety Quiz & Inorganic Lab Report Submission



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Make-up Lab Session

- For request of make-up session, please contact the lab officers or the lecturer-in-charge of each part.

- **Only If** medical certificate (MC), positive ART test result, NS document and university-related activities with strong reason(s) is provided.

- The proposed make-up session will be arranged by following the University's and Department's regulation for the face-to-face classes.

Assessments - Inorganic

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Note: Penalty will be applied for any late submission of risk assessment, lab report and proforma.

Inorganic part carries 50% for CM3191 as listed below.
[Note: For organic part, please contact Dr. Zhang for the details.]

1) Individual Risk Assessment for inorganic experiment 2 (Session II) [deadline: 10:00 a.m. on 16-Sep-2024 *via* Canvas-Assignments] (**5%**).
The file name of your risk assessment is your matric. number or full name (same as the name printed student ID card).

2) Closed-book Written Test for both organic and inorganic parts (**15%** for inorganic part only)

3) Individual "handwritten" proforma for inorganic experiment 1 (**15%**)

- The "ONE-file" softcopy handwritten proforma in "pdf" format (attached with datasheets, spectra and appendices, if any) must be uploaded into "Expt. 1 Proforma submission" via Canvas-Assignments by the deadline (10:00 a.m. on 8-Nov-2024, Fri.). Both submission *via* email and the hardcopy of lab report will NOT be accepted. The file name of your lab report is your matric. number or full name (same as the name printed student ID card).

4) Individual Short Lab Report for inorganic experiment 2 (**15%**)

- The lab report includes: Title, Aims, Result & Discussion, and data analysis and spectral interpretation, Conclusion, References. The maximum no. of page for SHORT report is **THREE (3) [A4 size]** for inorganic experiment 2. The figures and graphs as well as appendix (signed raw data & spectra) are excluded in this page limit. The "ONE-file" softcopy lab report in "pdf" format must be uploaded into "Expt. 2 short lab report submission" via Canvas-Assignments by the deadline (10:00 a.m. on 8-Nov-2024, Fri.). Both submission *via* email and the hardcopy of lab report will NOT be accepted. The file name of your lab report is your matric. number or full name (same as the name printed student ID card).

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Closed-book Written test, inorganic + organic (**15%** of CA for inorganic part only)

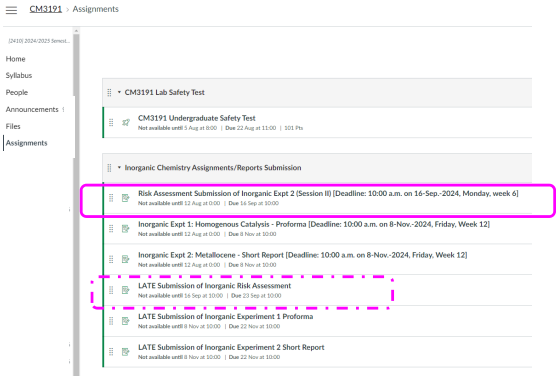
- Test the understanding of related theoretical and practical, safety aspects of both inorganic and organic experiments covered in this module including the practical / lectures / lecture notes
- Date: 11-Nov.-2024 (week 13, Monday.)
- Venue: LT26
- Time: during lecture slots
- Written Test Examiners this semester:
  Dr. Chan (Inorganic part);
  Dr. Zhang (Organic part)

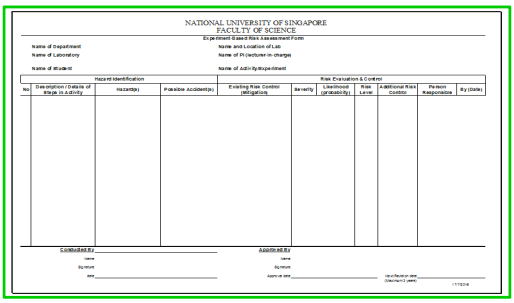
Note: For organic part, please check with Dr. Zhang.

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Individual Risk Assessment for Inorganic Expt. 2 (session II) – 5% of CA

- **The risk assessment form is available in Canvas-Files.**
- download the blank teaching lab experiment risk assessment form – **CM3191** experiments.xls
- submit your filled risk assessment for **inorganic expt. 2 (session II)** by **10:00 a.m. on 16-Sep-2024 (Mon.; week 6) via Canvas-Assignments** [refer to inorganic notes p.17]. **Penalty will be applied for any late submission.**
- The file name of your lab report is your matric. no. or full name (same as student ID).

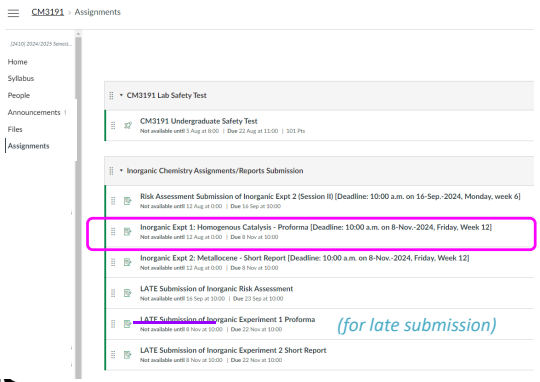




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Individual "handwritten" proforma for inorganic experiment 1 (15% of CA)

- The proforma form is available in Canvas-Files.
- The "ONE-file" softcopy handwritten proforma in "pdf" format (attached with datasheets, spectra and appendices, if any) must be uploaded into "Expt. 1 Proforma submission" via Canvas-Assignments by the deadline (10:00 a.m. on 8-Nov-2024, Fri.; week 12)
- Both submission *via* email and the hardcopy of lab report will NOT be accepted. The file name of your lab report is your matric. number or full name (same as the name printed student ID card).



- Requirements NMR spectra attached for expt. 1 proforma: Convert your NMR raw data (sent by GTA via your NUS email address) into a spectra (at least one FULL spectrum for each session) in which *file name must be shown*. One of the spectra must be full spectrum **from 0 ppm to 10 ppm** with the integration of **ALL signals** present on your spectra for each sessions II and III. All these data must be shown clearly. This is a part of lab-report assessment.

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Inorganic expt. 2 Lab Report Format: Individual Short Lab Report (max. no. of A4 pages: 3) – 15% of CA

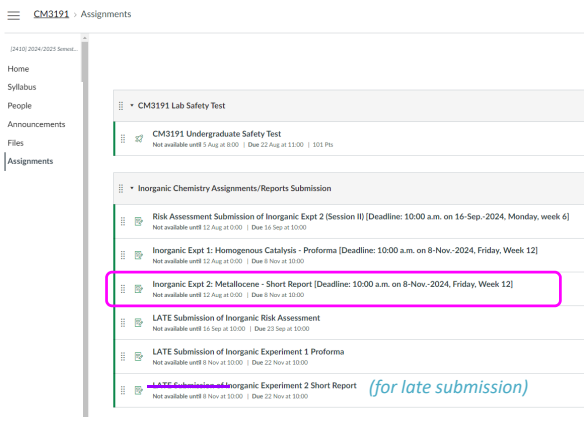
- **The file name of your lab report is your matric. no. or full name (same as student ID).**

Submission Deadline (via Canvas-Assignments) are:
ALL Groups –10:00 a.m. on 8-Nov.-2024 (Fri.; week 12)

Contents of short lab report:

- Your full name (as listed in your student ID)
- Your group session and fumehood number
- Title of experiment
- Aim of experiment (bullet form)
- Results and Discussion (with spectral and data analyses)
- Conclusion
- Reference
- Appendices (data sheets for raw data/observation, spectra, graphs, pictures taken in magnetic susceptibility measurement, microscopic images, calculation, and etc.)

[Note: Appendices will not be counted into page limit. **Penalty will be applied for any late submission**.]



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Details for the Inorganic expt. 2 Short Lab Report Requirements

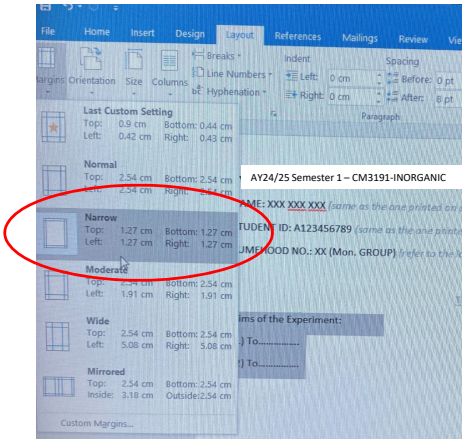
- This short report is for the entire inorganic experiment 2 (including all TWO sessions of inorganic experiment 2). For report submission, **PDF** format will be accepted in Canvas-Assignments, **but it must be a SINGLE file containing all the report contents, datasheet(s), and the spectra obtained.**

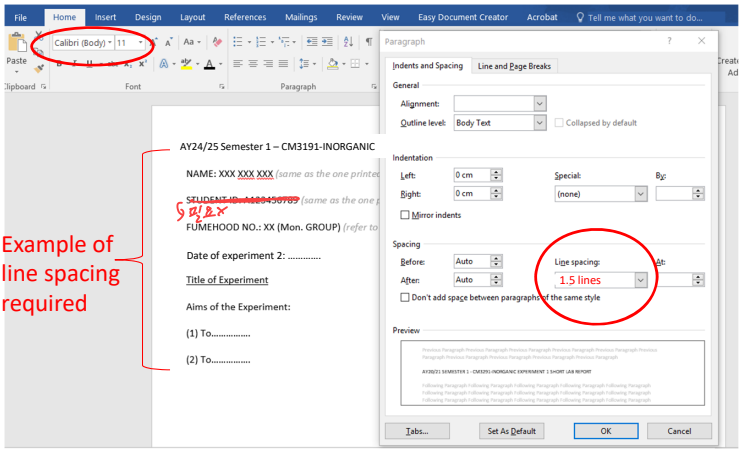
- Requirements NMR spectra attached for expt. 2: Convert your NMR raw data (sent by GTA via your NUS email address) into a spectra (at least one FULL spectrum) in which *file name must be shown*. One of the spectra must be full spectrum **from 0 ppm to 10 ppm** with the integration of **ALL signals** present on your spectrum. All these data must be shown clearly. This is a part of lab-report assessment.

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Details for the Inorganic expt. 2 Short Lab Report Requirements

- When you use Microsoft Word to prepare short lab report, the font size required is "**11**" and the font can be "**either Arial or Calibri**". Margins of the layout is "**Narrow**" (i.e. 1.27 cm for all margins including Top, bottoms, Left and Right in A4 page). In the setting of paragraph, the Line spacing for the entire report is "**1.5 lines**". Refer to the pictures below as reference. If you use other software to prepare the report, please follow the similar measurements or setting.





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Penalty for Late Submission of Lab report/Proforma/Risk Assessment

- Late submission results in a penalty of FIVE marks being deducted per day.
- fails to submit the report/proforma/risk assessment within 6 days after the deadline given → ZERO marks!

Additional Notes for hands-on experiments

- incompletion of the experiment → no report submission → ZERO marks!
- completion of the experiment → but fails to submit the report → ZERO marks!

Plagiarism Check

Refer to https://studentconduct.nus.edu.sg/wp-content/uploads/NUS-Code-of-Student-Conduct.pdf

- "The University takes a serious view of plagiarism. Any student found to have committed or aided and abetted the offence of plagiarism may be subject to disciplinary actions in accordance with the Section 3 (L) of Statute 06 (Discipline) of the National University of Singapore at:
  https://share.nus.edu.sg/registrar/info/statutes/Statute06.pdf

- In addition, the student may receive no mark/grade for the relevant academic assignment, project, or thesis; and he/she may fail or be denied a grade for the relevant subject or module. Students should adopt this rule:

  "You have the obligation to make clear to the assessor which is your own work, and which is the work of others. Otherwise, your assessor is entitled to assume that everything being presented for assessment is being presented as entirely your own work." "

  [Adapted from http://www.eng.nus.edu.sg/ugrad/SI_plagiarism.html / http://www.sicc3.science.nus.edu.sg/undergraduates/activities/discipline_offence.html]

- **Don't share your spectrum (and other data) to others, if noted, both of you are treated as cheating for that experiment → ZERO marks and such lab report will be kept by department for record! You may also receive an official warning letter from University Student Unit.**

Tips to avoid plagiarism

- originality of your report and prepare it individually
- do not copy from seniors' report(s)
- do not pass/sell your graded report(s) to your friends or juniors or others
- keep all the softcopies of your lab reports in a safe place

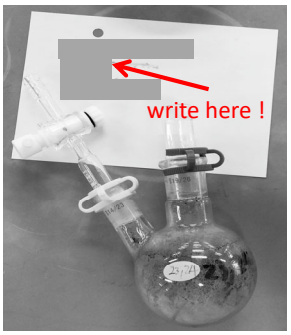
Please also be reminded that academic honesty is a MUST for the data/observation records and the respective report writing. For more details, please visit https://studentconduct.nus.edu.sg/wp-content/uploads/NUS-Code-of-Student-Conduct.pdf.

Notes for Sample Submission -1

- label your samples with your name, original FH no. assigned and Session (i.e. Mon or Thu), and the S/N number (if any) assigned by lab officers
- place your labeled samples (i.e. CM3191, Experiment 1, Mon, Sample Name, etc.) to the respective sample submission point







0.1961 $RuCl_2(PPh_3)_3$ 0.1961_g 0.2161_g

$2\,RuCl_3 + 3PPh_3 \rightarrow 2\,RuCl_2(PPh_3)_3 + 2Cl^- + OPPh_3$

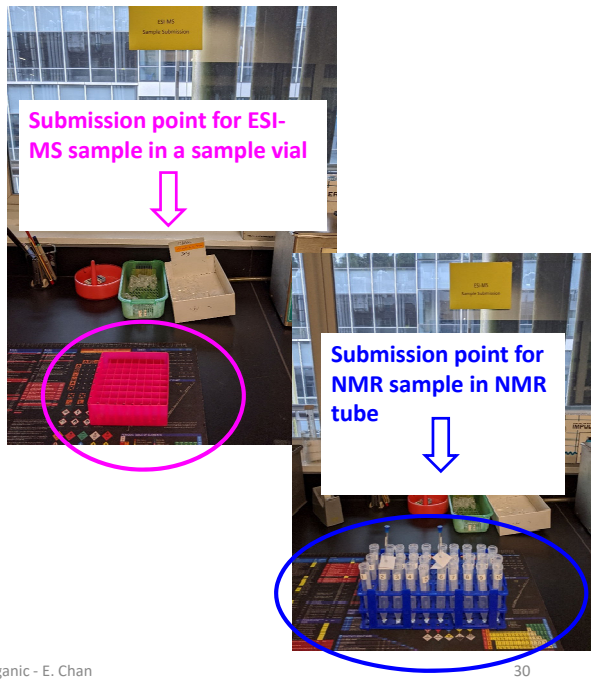
$\frac{2.0452\times10^{-4}}{3.9194\times10^{-4}\,mol}$ 958.83

Notes for Sample Submission - 2

- applicable to ~~EI-MS~~ ESI-MS and NMR sample submissions
- write your name and original FH No. assigned and lab session (Mon or Thu) on the white sticker of the sample tubes, and **the S/N number assigned by lab officers**







Any Risk from Labware ?

- glass, plastic, metallic labware with risks and hazards
- Examples:





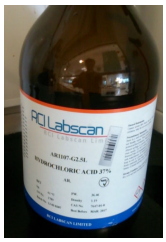
Glass capillary for TLC



Unused hot hotplate

Any Risk from liquid?

- assume all are potential risks and hazardous to health







Concentrated strong acids & bases

Highly Corrosive

Volatile or non-volatile



Flammable Solvents

Irritating Vapors

Incompatible with oxidizers

- halogenated & non-halogenated organics → designated waste bottles

Any Risk?

- Avoid any direct inhalation and physical contact !





Silica gel for column chromatography



Material Safety Data Sheet (MSDS) / Safety Data Sheet (SDS)

- MSDS for each inorganic experiment is available in "Canvas-Files-Inorganic Part".

Example 1

Praxair Material Safety Data Sheet

1. Chemical Product and Company Identification

Product Name: Nitrogen, refrigerated liquid	**Trade Names:** Liquid Nitrogen, Medipure® Liquid Nitrogen
Chemical Name: Nitrogen	**Synonyms:** Nitrogen (cryogenic liquid)
Chemical Family: Cryogenic liquid	**Product Grades:** Industrial, 5.0 Medical Drug

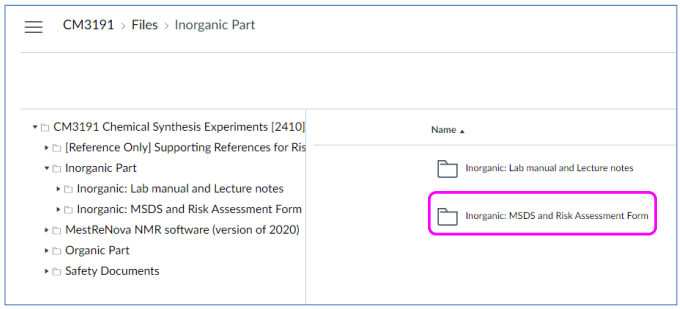
Telephone: **Emergencies:** 1-800-645-4633* **Company Name:** Praxair, Inc.
CHEMTREC: 1-800-424-9300* 39 Old Ridgebury Road
Routine: 1-800-PRAXAIR Danbury, CT 06810-5113

**Call emergency numbers 24 hours a day only for spills, leaks, fire, exposure, or accidents involving this product. For routine information, contact your supplier, Praxair sales representative, or call 1-800-PRAXAIR (1-800-772-9247).*

2. Hazards Identification

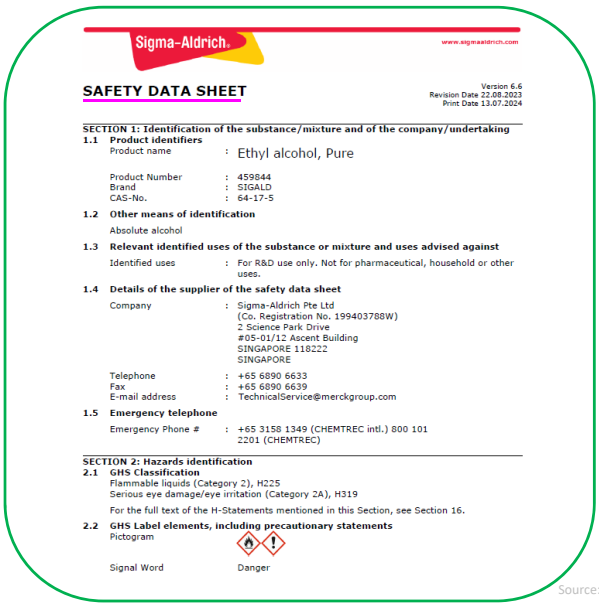
EMERGENCY OVERVIEW

WARNING! Extremely cold liquid and gas under pressure.
Can cause rapid suffocation.
Can cause severe frostbite.
May cause dizziness and drowsiness.
Self-contained breathing apparatus and protective clothing may be required by rescue workers.
Under ambient conditions, this is a colorless, odorless, cryogenic liquid.



Material Safety Data Sheet (MSDS) / Safety Data Sheet (SDS)

Example 2



Sigma-Aldrich.

www.sigmaaldrich.com

SAFETY DATA SHEET

Version 6.6
Revision Date 22.08.2023
Print Date 13.07.2024

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifiers

Product name : Ethyl alcohol, Pure

Product Number : 459844
Brand : SIGALD
CAS-No. : 64-17-5

1.2 Other means of identification

Absolute alcohol

1.3 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : For R&D use only. Not for pharmaceutical, household or other uses.

1.4 Details of the supplier of the safety data sheet

Company : Sigma-Aldrich Pte Ltd
(Co. Registration No. 199403788W)
2 Science Park Drive
#05-01/12 Ascent Building
SINGAPORE 118222
SINGAPORE

Telephone : +65 6890 6633
Fax : +65 6890 6639
E-mail address : TechnicalService@merckgroup.com

1.5 Emergency telephone

Emergency Phone # : +65 3158 1349 (CHEMTREC intl.) 800 101 2201 (CHEMTREC)

SECTION 2: Hazards identification

2.1 GHS Classification

Flammable liquids (Category 2), H225
Serious eye damage/eye irritation (Category 2A), H319

For the full text of the H-Statements mentioned in this Section, see Section 16.

2.2 GHS Label elements, including precautionary statements

Pictogram

Signal Word Danger

Source: https://www.sigmaaldrich.com/SG/

Emergency Equipment

- Do inform us in case you need or use it

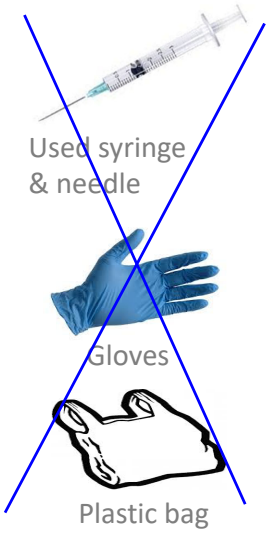








Proper Use of Broken Glass Bin

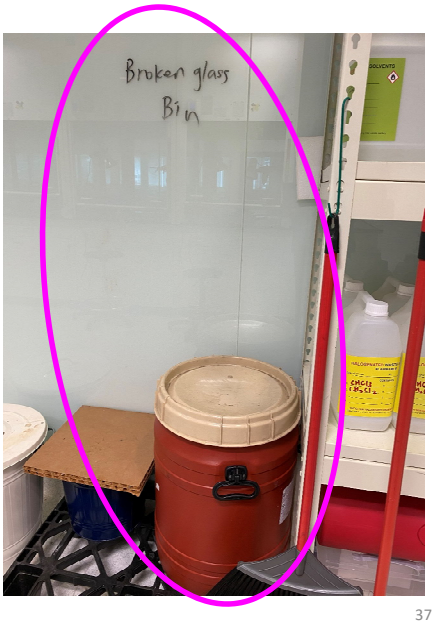


Broken glass





Or other type of Broken Glass Bin in the lab



Proper glassware cleaning after experiment ??

Good housekeeping ??







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General Introduction: *d*-block metals ≠ transition metal (only group 3–11)

1-2	3	4	5	6	7	8	9	10	11	12	13-18
***s*-block**	***d*-block**										***p*-block**
	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	
	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	
	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	

→ Metals of the 2nd and 3rd rows are sometimes called the "heavier *d*-block metals".

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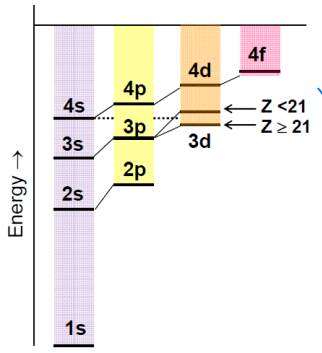
Source: CM3111 lecture notes by Dr. Edith Chan.

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Electronic Configurations for *d*-block Elements

only for element! not ion etc

- For elements with an incompletely filled d subshell, the determination of actual ground states by spectroscopy and calculation shows that it is advantageous to occupy orbitals predicted to be higher in energy (i.e. 4*s* orbitals for first row transition metals).
- Thus the ground state of these atom are of the form of $3d^n4s^2$, with fully occupied 4*s* orbitals (which is at higher energy) → reduce the repulsion among the electrons that would occur if the lower-energy 3*d* orbitals were occupied.
- In some cases, a lower total energy may be obtained by moving an *s* electron into the *d* subshells to form a half-filled or filled *d* subshell.
  e.g. Cu: $[Ar]3d^{10}4s^1$ (NOT $3d^94s^2$)



A schematic diagram of the energy of a many-electron atom with Z<21 (as far as Ca). There is a change in order for Z≥21 (from Sc onwards).

for e⁻ filling rule diff dep on Z.

Orbitals Filled			
1s			
2s			2p
3s			3p
4s		3d	4p
5s		4d	5p
6s	4f	5d	6p
7s	5f	6d	7p

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Electronic Configurations for *d*-block Cations or Complexes

- For the cations or complexes of (first row) d-block elements, the removal of electrons reduced the complicating effects of electron-electron repulsion and the 3*d* orbital energies fall well below that of the 4*s* orbitals. As a result, all d-block cations or complexes have d^n configuration without having electrons in the outermost *s* orbitals.

- Examples
  (i) Fe: [Ar]$3d^6 4s^2$
  (ii) Fe^{2+} : [Ar]$3d^6$
  (iii) Fe^{3+} : [Ar]$3d^5$
  (iv) $[Fe(CO)_5]$: [Ar]$3d^8$

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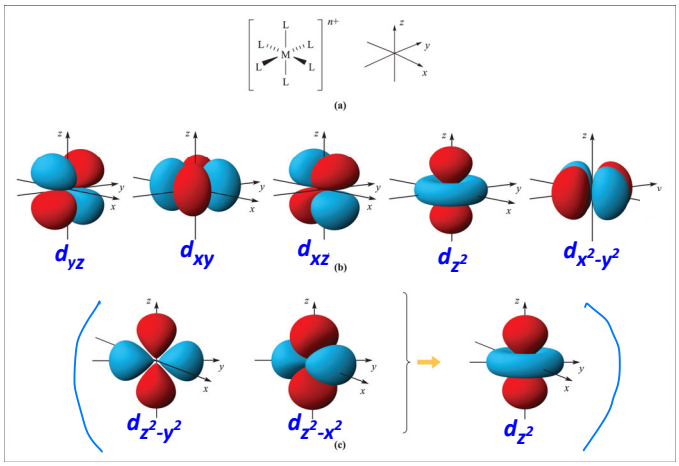
Crystal Field Theory

- An approach to the bonding in complexes of the *d*-block metals;
- An electrostatic model;
- Use the ligand electrons to create an electric field around the metal centre;
- Consider ligands as point charges without metal-ligand covalent interactions.

some have covalent.
this rule is just 'strict' def

Source: CM3111 lecture notes by Dr. Edith Chan.

Five *d* orbitals in Octahedral (O_h) Complex



(a) The six M–L vectors of an octahedral complex $[ML_6]^{n+}$ can be defined to lie along the *x*, *y* and *z* axes.

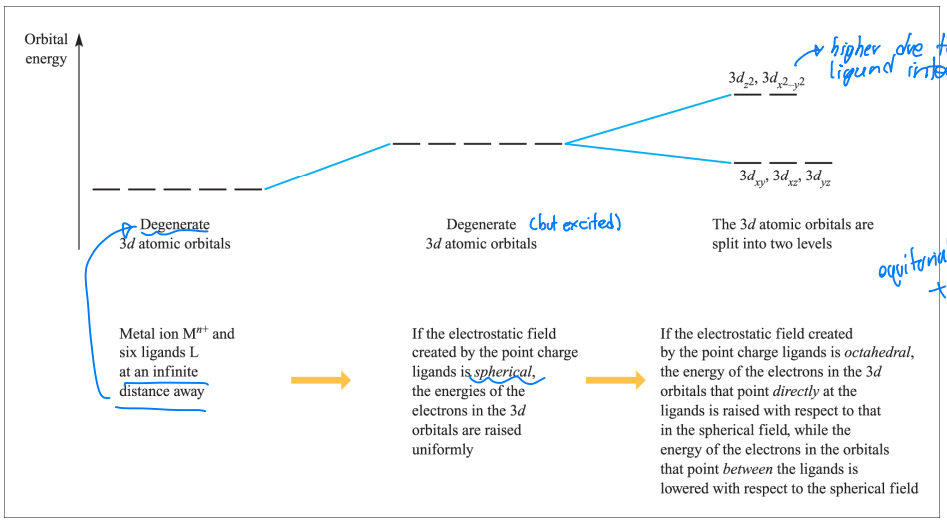
(b) The five *d* orbitals; the d_z2 and $d_x2_{-y}2$ atomic orbitals point directly along the axes, but the d_{xy}, d_{yz} and d_{xz} atomic orbitals point between them.

(c) The formation of a d_z2 orbital from a linear combination of $d_z2_{-y}2$ and $d_z2_{-x}2$ orbitals.

Reference: C. E. Housecroft, A. G. Sharpe, *Inorganic Chemistry*, 4th ed., Pearson.

Source: CM3111 lecture notes by Dr. Edith Chan.

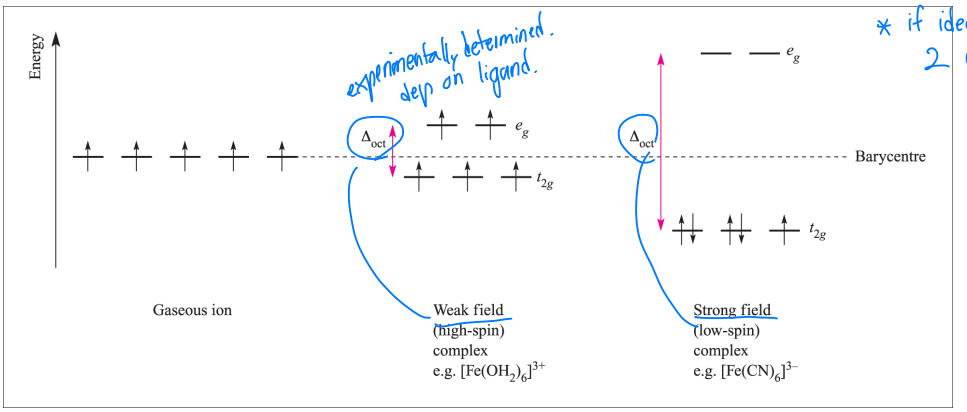
The Octahedral Crystal Field



Reference: C. E. Housecroft, A. G. Sharpe, *Inorganic Chemistry*, 4th ed., Pearson.

Source: CM3111 lecture notes by Dr. Edith Chan.

The Octahedral Crystal Field



- The occupation of the 3*d* orbitals in weak and strong field Fe^{3+} (d^5) complexes.
- Δ_{oct} is an experimental quantity.

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Stereochemistry of metal complexes or coordination compounds

(i) electrostatic factor

- ligand repulsions that tend to stabilize the most energetically favorable geometry

(ii) electronic factor

- stability of each possible geometry for a given electron configuration (d^n) of transition metals

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Ligands

- From the electronic point of view, ligands are molecules or ions with atoms from the non-metallic elements, which act as electron donors.
- The most common atoms are N, P, O, S, X (halogens) and C.

• Roles:

(i) influence the central metal ion significantly;

(ii) can display reactivity and undergo chemistry of their own while still bound to the metal ion, without any change in the fundamentals of their behaviors as ligands.

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Nomenclature of Anionic and Neutral Ligands

- **Anionic ligand**:

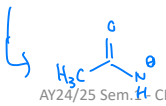
Upon coordination to a metal, ending changes from "*-e*" to "*-o*".
e.g. "-ide", "-ite" and "-ate" → "-ido", '-ito' and "-ato"

Certain simple ligands have historically been represented in names by abbreviated form: fluoro, chloro, iodo, hydroxo, hydro, cyano, etc. Following the rule stated above, they are fluorido, chlorido, etc. (Both forms can be found in literature)

- **Neutral ligand:**

used without modification. (But coordinated water is referred to as aqua; coordinated ammonia is referred to as ammine.)
e.g. $MeCONH_2$ = acetamide (not acetamido)

deprotonated anionic.



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INORGANIC Experiment 1

Homogenous Catalysis: Tris(triphenylphosphine)ruthenium(II) Dichloride – Synthesis, Characterization and Catalytic Studies

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$

Session II: Controlled Study - Oxidation of 1-phenylethanol to acetophenone in the absence of catalyst

Session III: Study of Catalytic Reaction: $RuCl_2(PPh_3)_3$–catalysed Oxidation of 1-phenylethanol to acetophenone

Disclaimer: All materials, information, weblinks and pictures contained in this video are for educational purposes only.

Ruthenium: oxidation states

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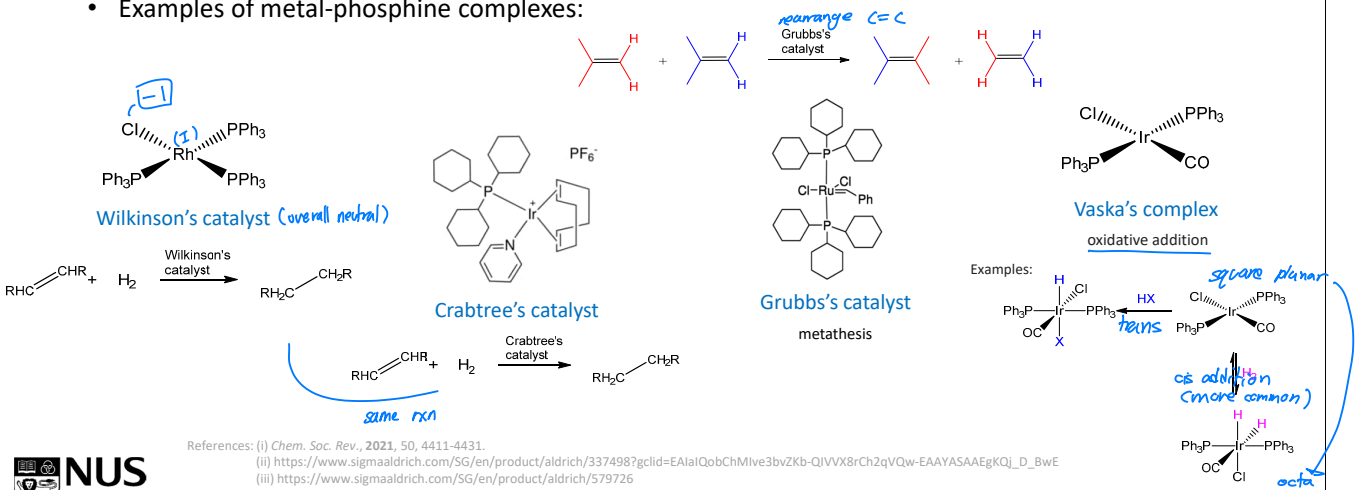
- Oxidation states (the most stable states are highlighted in blue):

Group	3	4	5	6	7	8	9	10	11	12
2nd row	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
				0	0	0	0	0		
					1		1		1	[1]
		2	2	2	[2]	2	2	2	2	2
	3	3	3	3	3	3	3		3	
		4	4	4	4	4	4	4		
			5	5	5	5	5			
				6	6	6	6			
					7	7				
						8				

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Phosphine Ligands

- Commonly used ligands in coordination chemistry and homogeneous transition metal catalysis;
- Examples of metal-phosphine complexes:



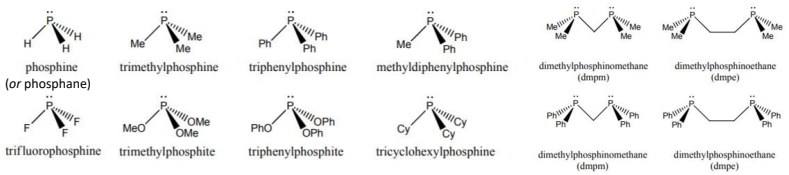
References: (i) *Chem. Soc. Rev.*, **2021**, 50, 4411-4431.
(ii) https://www.sigmaaldrich.com/SG/en/product/aldrich/337498?gclid=EAIaIQobChMIve3bvZKb-QIVVX8rCh2qVQw-EAAYASAAEgKQj_D_BwE
(iii) https://www.sigmaaldrich.com/SG/en/product/aldrich/579726

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Tertiary Phosphine Ligands

- Tertiary phosphines, PR_3 (which are also named as tertiary organophosphanes) are an important class of ligands because their electronic and steric properties can be varied by changing the R group(s).

Examples:



- Similar to NR_3, phosphines have a lone pair on the central atom that can be donated to a metal *via* σ-interaction. But PR_3 can act as π-acceptors to an extent that depends on the nature of the R groups present.
- For CO, the π* orbital accepts electrons from the metal. However, in PR_3, the σ* orbitals of the P–R bonds play the role of acceptor as reported in some sources and there is also a current agreement more appropriate given the relatively high energy of a phosphorous *d*-orbital.
- The final order of increasing π-accepting properties:

$$PMe_3 < PPh_3 < P(OMe)_3 < P(OPh)_3 < PCl_3 < PF_3 \approx CO$$

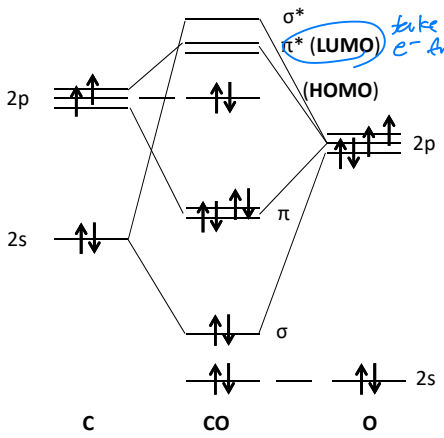
Sources:
(i) http://alpha.chem.umb.edu/chemistry/
(ii) (ii) C. E. Housecroft, A. G. Sharpe, *Inorganic Chemistry*, 4th ed., Pearson, **2012**.

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Supplementary

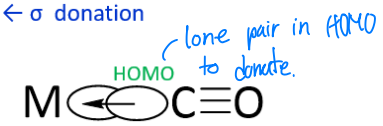
Note: The details on π-back bonding and molecular orbital theory are covered in CM3111.

MO diagram for carbon monoxide

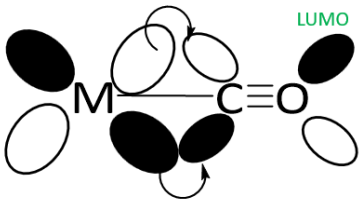


Reference: Alan K. Brisdon, *Inorganic Spectroscopic Methods*, Oxford University Press, 2001.

(i) dσ ← σ donation



(ii) dπ → π* donation (π-back bonding)

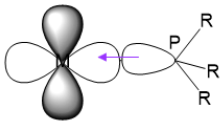


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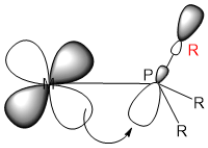
Supplementary

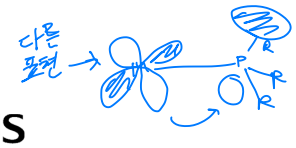
Note: The details on π-back bonding and molecular orbital theory are covered in CM3111.

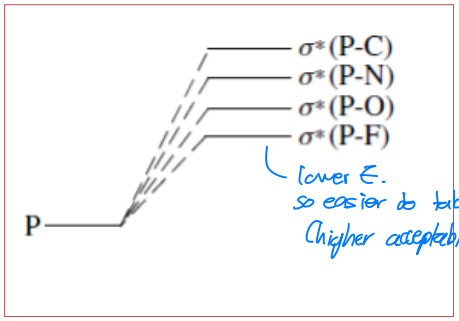
(i) dσ ← σ donation



(ii) dπ → σ* donation (π-back bonding)







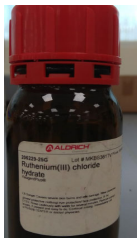
- The final order of increasing π-accepting properties:

$PMe_3 < PPh_3 < P(OMe)_3 < P(OPh)_3 < PCl_3 < PF_3 \approx CO$

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trans-$[RuCl_2(PPh_3)_3]$: Synthesis

- Starting materials:

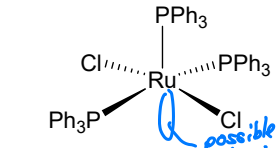


$RuCl_3 . xH_2O$



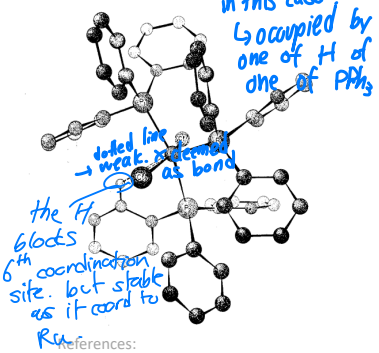
PPh_3

not only act as ligand. oxidises.

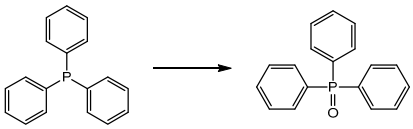


possible coord site but is this empty in this case? ↳ occupied by one of H of one of PPh_3

Molecular structure of $RuCl_2(PPh_3)_3$ in solid state



- In the synthesis, redox
  - (i) Reduction: $Ru^{III} \rightarrow Ru^{II}$
  - (ii) Oxidation:



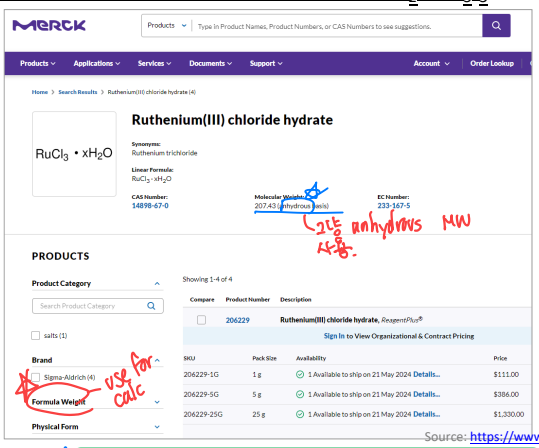
References:

(i) *J. Chem. Edu.* **2011**, 88, 657-661.
(ii) *J. Chem. Educ.* **2017**, 94, 928 - 931.
(iii) *Inorg. Chem.*, **1965**, 778–783

*H_2O can act as aqua ligand → 为了 minimise contact in H_2O

Inorganic experiment 1: Homogenous Catalysis

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$



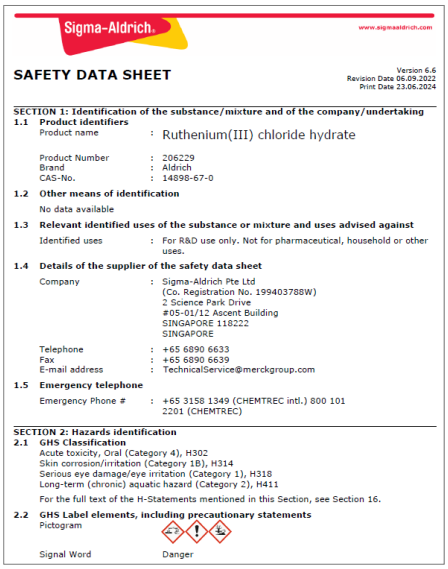
anhydrous MW

* Data sheet 要把 all raw data write down.

use for calc

Source: https://www.sigmaaldrich.com





https://www.sigmaaldrich.com/SG/en/sds/aldrich/206229

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Supplementary

TGA

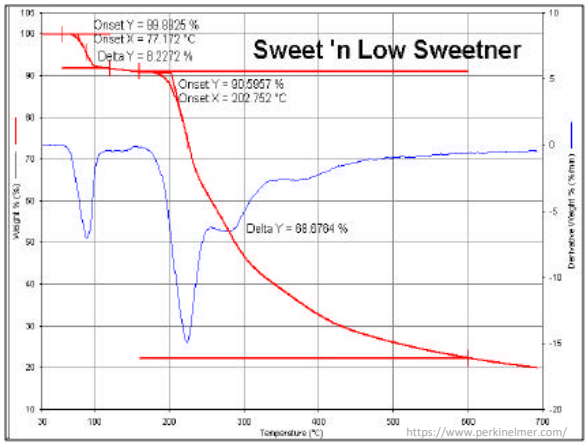
to identify mol of H_2O per mol of starting mat.

$RuCl_3 \cdot xH_2O$



Model: Pyris 6 TGA

- The method to determine the moisture content of the sample is thermogravimetric analysis (TGA).
- TGA measures the properties of mass loss properties and the rate of weight loss as a function of time or temperature.

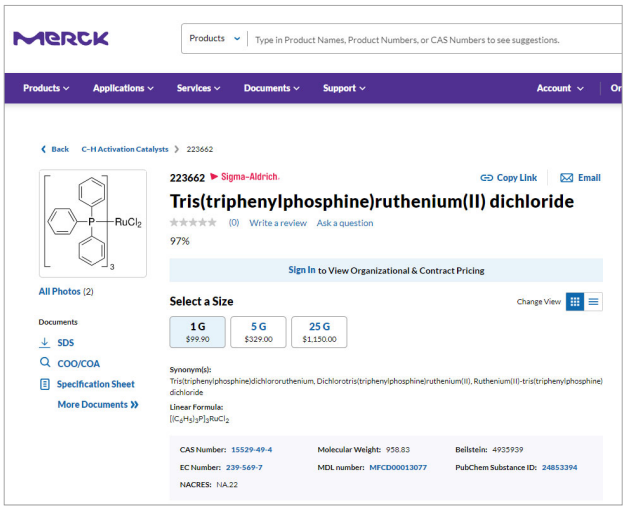


Example: TGA results obtained for Sweet 'n Low (artificial) sweetener

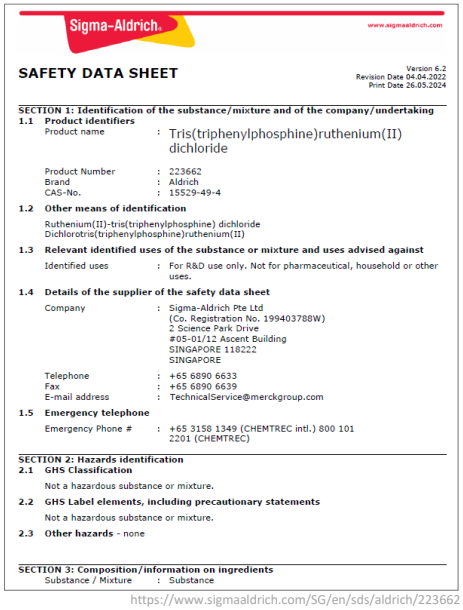
Source: https://www.perkinelmer.com/

Inorganic experiment 1: Homogenous Catalysis

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$



Source: https://www.sigmaaldrich.com



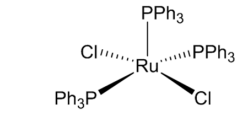
https://www.sigmaaldrich.com/SG/en/sds/aldrich/223662

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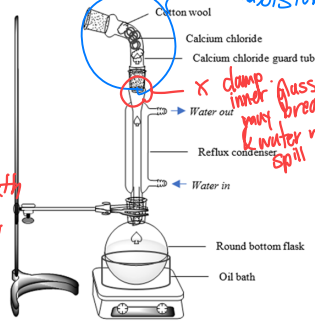
Inorganic experiment 1: Homogenous Catalysis

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$

1. Weigh 0.50 g of PPh_3 and transfer it into a 100 mL round bottom flask fitted with a stirrer. Then add 20 mL of dried absolute ethanol into the flask. Warm the solution (~ 40 °C) for 10 - 15 minutes to allow the solids to dissolve.
2. Weigh 0.08 g of $RuCl_3.xH_2O$ and transfer it into a 50 mL round bottom flask fitted with a stirrer bar. Then add 20 mL of dried absolute ethanol into the flask. Allow the $RuCl_3.xH_2O$ solid to dissolve *via* constant stirring using the stirrer bar.
3. Using a glass dropper, transfer the ethanolic solution containing the dissolved $RuCl_3.xH_2O$ into the 100 mL round bottom flask containing the dissolved PPh_3 in ethanol. Add 5 mL of dried absolute ethanol to rinse all $RuCl_3.xH_2O$ residue into the reaction flask containing PPh_3.
4. Connect a water-cooled reflux condenser with a calcium chloride drying tube to the 100 mL round bottom flask containing the reactants [*See* ***Figure 5***].
5. Reflux the reaction mixture for **90 minutes**. *[Note: the temperature of oil bath is < 130 °C.]*



Molecular structure of $RuCl_2(PPh_3)_3$ in solid state



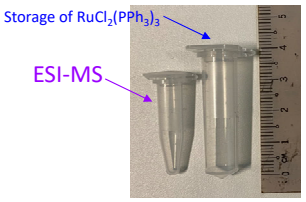
Apparatus used in the synthesis of $RuCl_2(PPh_3)_3$

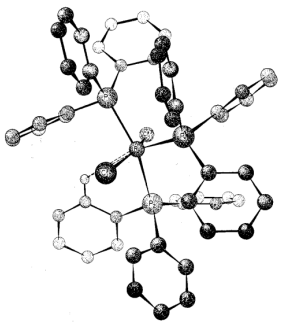
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Inorganic experiment 1: Homogenous Catalysis

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$

6. Allow the reaction mixture to cool to room temperature after the reflux.
7. Using a Hirsch funnel, collect the solid, $RuCl_2(PPh_3)_3$ by suction filtration. Wash the solid with cold dried absolute ethanol (2 x 3 mL) and then dry it under vacuum for 20 – 30 minutes.
8. Weigh the amount of dried $RuCl_2(PPh_3)_3$ obtained and record it on datasheet.
   [Note: The detailed calculation of % yield of $RuCl_2(PPh_3)_3$ obtained must be shown in the lab report.]
9. Transfer the dried $RuCl_2(PPh_3)_3$ obtained (~ 10 mg) into a sample tube. Label the sample tube clearly and submit it for Electrospray Ionisation Mass Spectrometry (ESI-MS) analysis.
10. Obtain the solid IR spectrum (range: 500–4000 cm^{-1}) of the dried $RuCl_2(PPh_3)_3$ obtained by *either* ATR-IR *or* KBr disc.
    [Note: The IR characterization method chosen must be recorded on the datasheet and in the lab report.]

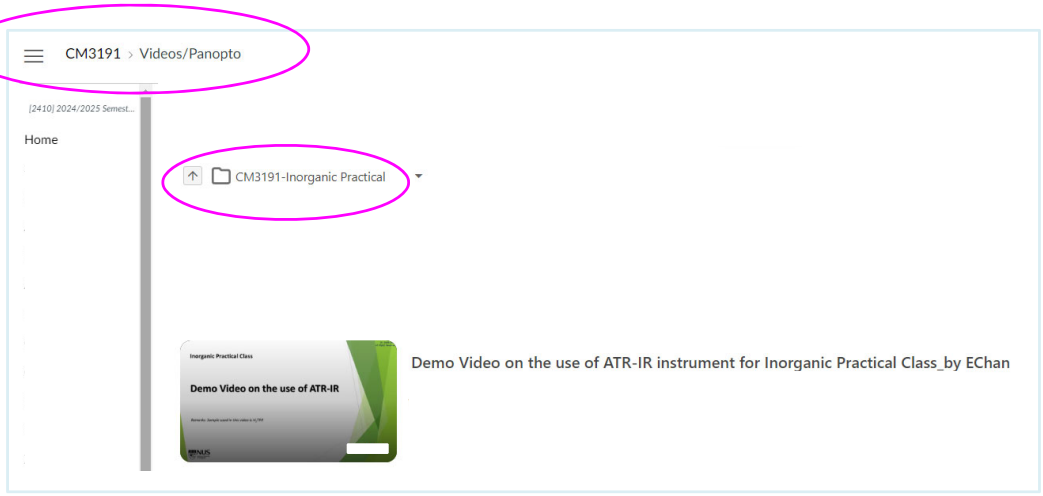




Inorg. Chem., **1965**, 778–783

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ATR-IR: Demo Video in Canvas-Videos/Panopto



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Inorganic experiment 1: Homogenous Catalysis

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$

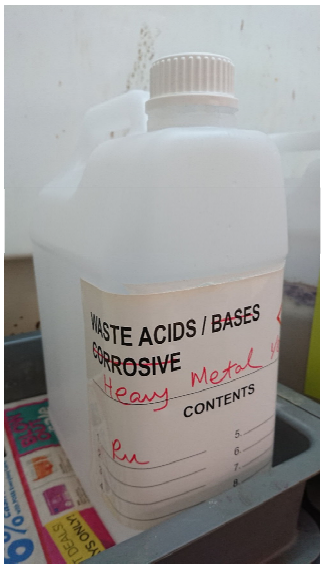
At the end of this session:

(i) Submit the sample vial (with clear and proper label) containing the dried $RuCl_2(PPh_3)_3$ for Electrospray Ionisation Mass Spectrometry (ESI-MS) analysis.

(ii) Record all the raw data and observation of the reaction on your own datasheet (signed by your GTA at the end of session).

(iii) Obtain the solid IR spectrum of the dried $RuCl_2(PPh_3)_3$ characterized by ATR-IR or KBr disc.

Ru waste bottle



Inorganic experiment 1 (session I): Risk assessment

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Experiment-Based Risk Assessment Form

Name of Department	Department of Chemistry	Location of Lab	S5-04
Name of Laboratory	Chemical Synthesis Teaching Lab	Name of PI (lecturer-in-charge)	Dr. Edith S. H. Chan
Names of LOs	CM3191 STUDENTS (one performing experiment)	Name of Activity/Experiment	CM3191 Inorganic - Homogenous Catalysis: Session I - Synthesis and Characterization of RuCl2(PPh3)3

Hazard Identification				Risk Evaluation and Control						
No	Description/Details of Steps in Activity	Hazards	Possible Accident / Ill Health & Persons-at-Risk	Existing Risk Control (Mitigation)	Severity	Likelihood (Probability)	Risk Level	Additional Risk Control	Person Responsible	By (Date)
1	Clean and dry all the glassware (round-bottomed flask, conical flasks, beakers, etc) used in the reactions	**Physical hazard**: Broken Glassware	Cuts from broken glassware	Visual inspection for any cracks before usage. Proper disposal of broken glassware in Sharps container. Handle & use glass dropper and glassware with care.	2	1	2		one performing exp. NOT TA etc	
2	Weigh 0.50 g of PPh3 and transfer it into a 100 mL round bottom flask fitted with a stirrer. Then add 20 mL of dried absolute ethanol into the flask. Warm the solution (~ 40 °C) for 10 - 15 minutes to allow the solids to dissolve.	**Chemical hazard**: PPh3 is an irritant and harmful when swallowed	Causes irritation to skin. Toxic when ingested. Repeated exposure can be toxic to the body. Spillage of triphenylphospine	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Ensure that reagent bottle containing PPh3 is capped after use and clean any spillage immediately	2	1	2			
		Chemical and fire hazard: Ethanol is both an irritant and flammable	Ethanol can cause eye irritation or damage. Ethanol is flammable. Spillage of ethanol during the transfer process	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep ethanol away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Heat hazard: Usage of oil bath and hotplate during reflux	Spillage of oil bath. Hot oil in oil bath and hot surface of hot plate can cause scalding of skin.	Ensure appartus is properly set-up and clamped and condenser is not leaking water into the oil bath. Turn off the hotplate after reflux is completed	2	1	2			

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3	Weigh 0.08 g of RuCl3.xH2O and transfer it into a 50 mL round bottom flask fitted with a stirrer bar. Then add 20 mL of dried absolute ethanol into the flask. Allow the RuCl3.xH2O solid to dissolve via constant stirring using the stirrer bar.	**Chemical hazard**: RuCl3.xH2O is an irritant and is corrosive	May cause severe skin or eye irritation, corrosion or damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical and fire hazard: Ethanol is both an irritant and flammable	Ethanol can cause eye irritation or damage. Ethanol is flammable Spillage of ethanol during the transfer process	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep ethanol away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
4	Using a glass dropper, transfer the ethanolic solution containing the dissolved RuCl3.xH2O into the 100 mL round bottom flask containing the dissolved PPh3 in ethanol. Add 5 mL of dried absolute ethanol to rinse all RuCl3.xH2O residue into the reaction flask containing PPh3.	**Chemical hazard**: RuCl3.xH2O is an irritant and is corrosive	Spillage of chemical during transferring. May cause severe skin or eye irritation, corrosion or damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical and fire hazard: Ethanol is both an irritant and flammable	Spillage of chemical during transferring. Ethanol can cause eye irritation or damage. Ethanol is flammable Spillage of ethanol during the transfer process	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep ethanol away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Chemical hazard: PPh3 is an irritant and harmful when swallowed	Spillage of chemical during transferring. Causes irritation to skin. Toxic when ingested. Repeated exposure can be toxic to the body Spillage of triphenylphospine	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Ensure that reagent bottle containing PPh3 is capped after use and clean any spillage immediately	1	1	1			
		Physical hazard: Broken glass dropper or glassware	Cuts from broken glassware	Visual inspection for any cracks before usage. Proper disposal of broken glassware in Sharps container. Handle & use glass dropper and glassware with care.	2	1	2			

lab manual set 5+01 → specify components in the mixture (+ product)
- since method of risk treatment may be diff (e.g. ethanol is flammable)

hazardous when stirring, heating

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5	Connect a water-cooled reflux condenser with a calcium chloride drying tube to the 100 mL round bottom flask containing the reactants (PPh3, RuCl3.xH2O and ethanol). Reflux the reaction mixture for 1.5 hour. Allow the reaction mixture to cool to room temperature after reflux is completed.	**Chemical hazard**: RuCl3.xH2O is an irritant and is corrosive	May cause severe skin or eye irritation, corrosion or damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical and fire hazard: Ethanol is both an irritant and flammable	Ethanol can cause eye irritation or damage. Ethanol is flammable Spillage of ethanol during the transfer process	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep ethanol away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Chemical hazard: PPh3 is irritant and harmful when swallowed	Causes irritation to skin. Toxic when ingested. Repeated exposure can be toxic to the body Spillage of triphenylphospine	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Ensure that reagent bottle containing PPh3 is capped after use and clean any spillage immediately	2	1	2			
		Physical hazard: Broken glass dropper or glassware	Cuts from broken glassware	Visual inspection for any cracks before usage. Proper disposal of broken glassware in Sharps container. Handle & use glass dropper and glassware with care.	2	1	2			
		Heat hazard: Usage of oil bath and hotplate during reflux	Spillage of oil bath. Hot oil in oil bath and hot surface of hot plate can cause scalding of skin.	Ensure appartus is properly set-up and clamped and condenser is not leaking water into the oil bath. Turn off the hotplate after reflux is completed	2	1	2			
6	Using a Hirsch funnel, collect the solid, RuCl2(PPh3)3 by suction filtration. Wash the solid with cold dried absolute ethanol (2 x 3mL) and then dry it under vacuum for 20 – 30 minutes.	**Chemical hazard**: RuCl2(PPh3)3 is harmful and an irritant	Harmful if swallowed. Cause skin and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment.	2	1	2			
		Mechanical & Physical hazards: Usage of set-up for suction filtration	Glassware might topple down when the vacuum is used and could break, resulting in cuts.	1. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 2. Visually inspect apparatus for cracks before use 3. Clamp the apparatus properly before usage	2	1	2			
		Chemical and fire hazard: Ethanol is both an irritant and flammable	Ethanol can cause eye irritation or damage. Ethanol is flammable Spillage of ethanol during the transfer process	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep ethanol away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
7	Transfer the dried RuCl2(PPh3)3 obtained (~ 10 mg) into a sample tube. Label the sample tube clearly and submit it for Electrospray Ionisation Mass Spectrometry (ESI-MS) analysis.	**Chemical hazard**: RuCl2(PPh3)3 is harmful and an irritant	Harmful if swallowed. Cause skin and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment.	2	1	2			

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8	Obtain the solid IR spectrum (range: 500–4000 cm-1) of the dried RuCl2(PPh3)3 obtained by either ATR-IR or KBr disc.	**Chemical hazard**: RuCl2(PPh3)3 is harmful and an irritant	Harmful if swallowed. Cause skin and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment.	2	1	2			
		Chemical hazard: KBr is harmful and an irritant	Harmful if swallowed. Cause skin and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment.	2	1	2			
		Electrical hazard: exposed wire of the cable and/or broken plug or socket for the spectrometer	Electric shock if used improperly	1. Check the cable, plug and socket before use. 2. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for	2	1	2			
		Radiation Hazard: IR radiation	Injury from IR beam	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment.	2	1	2			
9	Washing of glasswares with acetone after experiment is completed	**Chemical and fire hazard:** Acetone is an irritant and is flammable	Can cause irritation to skin and eyes. Flammable substance that can start a fire.	1.Wash the apparatus in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Avoid placing acteone near any open or naked flame 4. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Physical hazard: Broken glasswares	Cuts and injury from broken glassware	1.Wash the apparatus in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Handle the glasswares with care to avoid dropping them 4. Dispose the broken glassware into the appropriate Sharps bin	2	1	2			

Conducted By
Name
Signature
Date

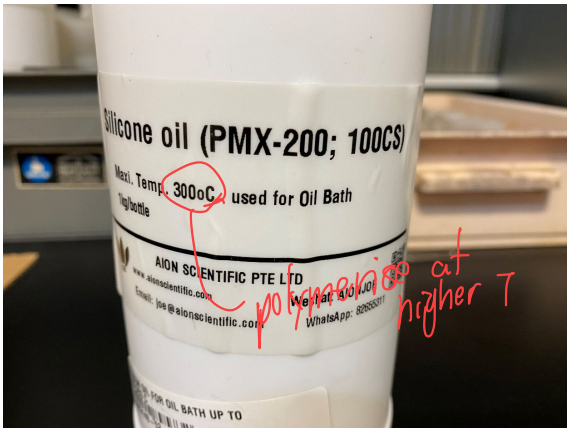
Approved By
Name Dr. Edith Chan
Signature
Approval date Next Revision date
(Maximum 3 years)

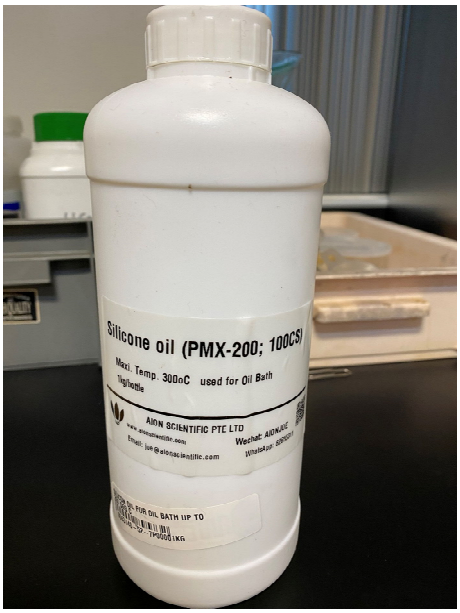
* for Ru complex, use DCM for washing. not acetone (not so effective)

* make sure IR spec is closed after putting sample.

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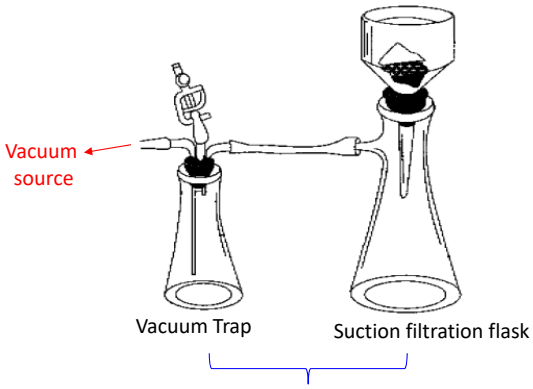
Silicone oil in oil bath



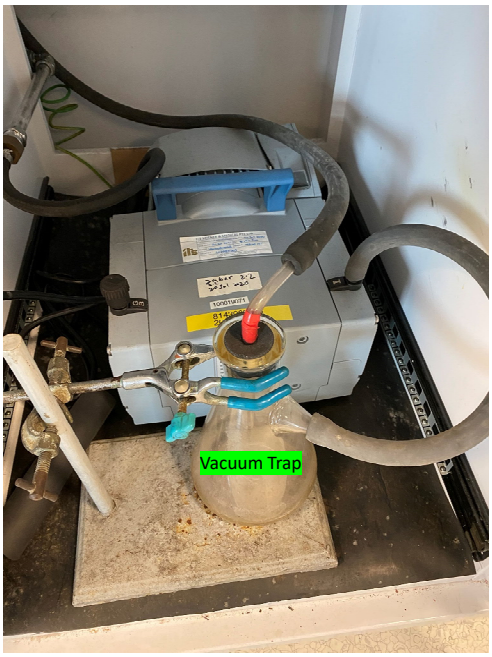


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Vacuum Trap



Both flasks must be clamped properly.



Inorganic experiment 1: Homogenous Catalysis

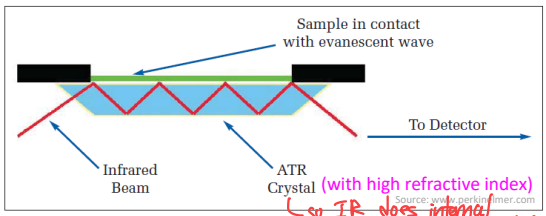
Session I: ATR-IR Characterization of $RuCl_2(PPh_3)_3$

Remarks: This technique will be also used in inorganic experiment 2.

Advantages of Attenuated Total Reflection (ATR):

- Little or no sample preparation, and easy to clean up
- Analysis of samples in their original states (no dilution with IR transparent salt)
- Work well with thick samples (e.g. polymer)

Graphical representation of a multiple reflection ATR:



An ATR accessory operates by measuring the changes that occur in a totally internally reflected infrared beam when the beam comes into contact with a sample.

(with high refractive index)
↳ so IR does internal reflection → operated in open air cambe
↕
KBr, need to close.

< ADDITIONAL TRAINING >

Use of Micropipettes



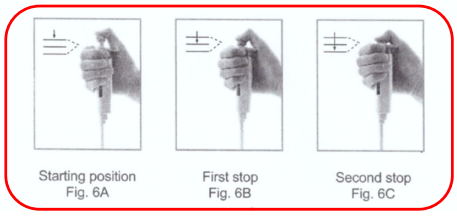
6. PIPETTING TECHNIQUES

6.1. Forward pipetting

Make sure that the tip is firmly attached to the tip cone. For best results the thumb button should be operated slowly and smoothly at all times, particularly with viscous liquids.

Hold the pipette vertically during aspiration. Make sure that the liquid and container vessel are clean and that the pipette, tips and the liquid are at the same temperature.

1. Depress the thumb button to the first stop (Fig.6B).
2. Place the tip just under the surface of the liquid (2-3mm) and smoothly release the thumb button. Carefully withdraw the tip from the liquid, touching against the edge of container to remove excess.
3. Liquid is dispensed by gently depressing the thumb button to the first stop (Fig.6B). After a short delay continue to depress the thumb button to the second stop (Fig.6C). This procedure will empty the tip and ensure accurate delivery.
4. Release the thumb button to the ready position (Fig.6A). If necessary change the tip and continue with pipetting.

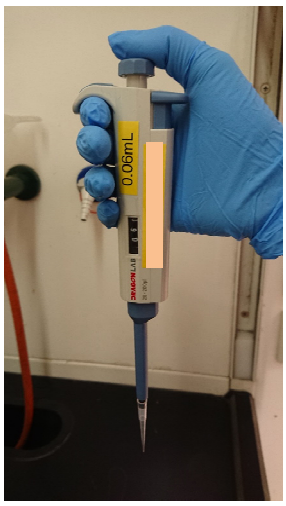


Starting position Fig. 6A | First stop Fig. 6B | Second stop Fig. 6C

< ADDITIONAL TRAINING >

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Use of Micropipettes

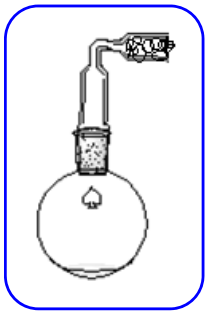




Inorganic experiment 1: Homogenous Catalysis

Session II: Controlled Study - Oxidation of 1-phenylethanol to acetophenone in the absence of Catalyst

1. Add 8 mL of dried acetone and a magnetic stirrer bar into a 100 mL round bottom flask.
2. Using a micropipette, add 0.06 mL of 1-phenylethanol into the reaction flask upon stirring.
3. Upon stirring, add 0.07 g of K_2CO_3 into the reaction flask.
4. Connect quickly a water-cooled reflux condenser with a calcium chloride drying tube to the reaction flask containing the reactants [*See* ***Figure 5***].
5. Reflux the reaction mixture for **60 minutes**. *[Note: The temperature of oil bath is < 100 °C.]*
6. Cool the reaction mixture to room temperature after the reflux.
7. Upon stirring, add 50 mL of *n*-hexane into the reaction flask to quench the reaction.



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Inorganic experiment 1: Homogenous Catalysis

Session II: Controlled Study - Oxidation of 1-phenylethanol to acetophenone in the absence of Catalyst

8. Using a Hirsch funnel, filter the reaction mixture through Celite by suction filtration and collect the filtrate. *[Note: The suction flask must be washed and dried completely before use.* ***When the suction is on, a thin layer of Celite should be packed on top of the filter paper in the Hirsch funnel.***
9. Transfer the filtrate from the suction flask into a 150 mL round bottom flask. Rinse the suction flask with *n*-hexane (2 x 5 mL) and transfer it into the round bottom flask.
10. Using a rotary evaporator, remove the solvent from the reaction mixture in *n*-hexane under reduced pressure.
11. Connect the round bottom flask with a gas adapter and dry the oily product under vacuum for 15 - 20 minutes.
12. Using a micropipette, add 0.07 mL of mesitylene into the reaction flask upon stirring.
13. Add 0.6 mL of $CDCl_3$ into the round bottom flask containing oily product and transfer all solution into an NMR tube with a clean glass dropper. Label the NMR tube clearly and submit it for ^1H NMR analysis.



Celite

